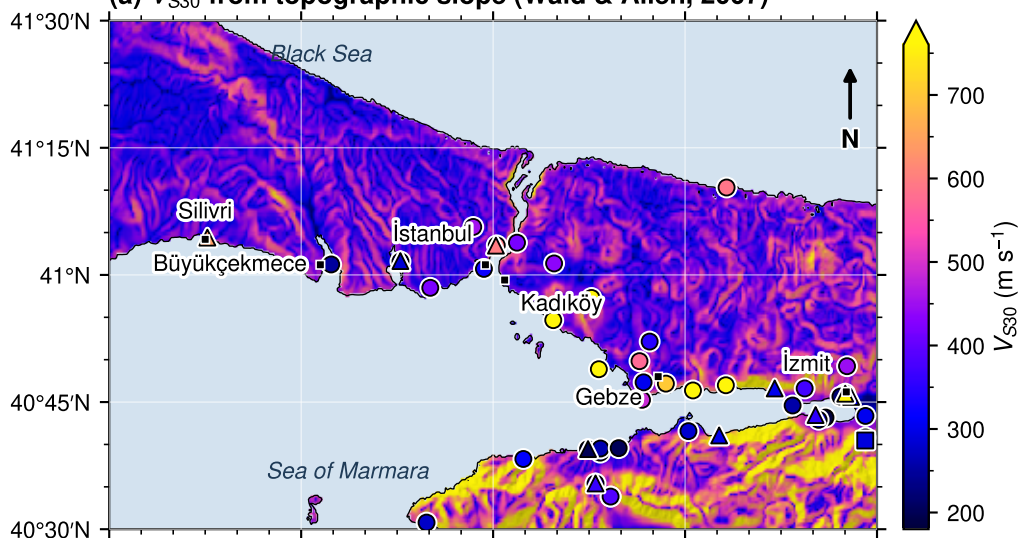
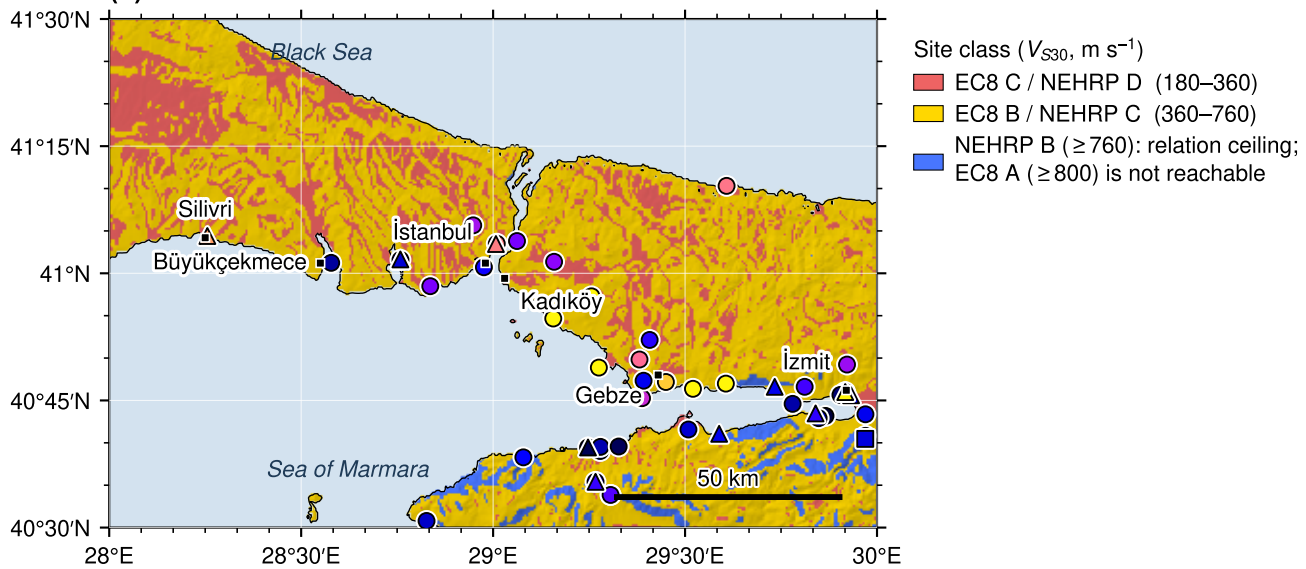


Site characterisation of the Istanbul metropolitan area

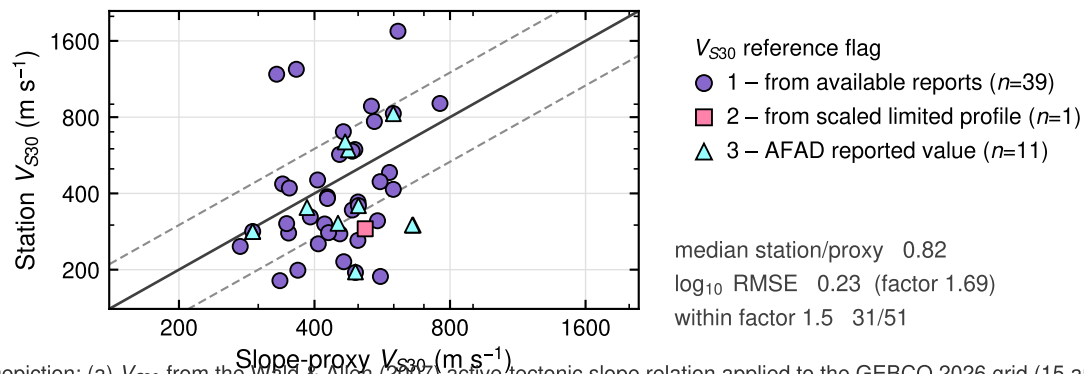
(a) V_{S30} from topographic slope (Wald & Allen, 2007)



(b) EC8 / NEHRP site classes



(c) Station V_{S30} against the slope proxy



Depiction: (a) V_{S30} from the Wald & Allen (2007) active tectonic slope relation applied to the GEBCO 2026 grid (15 arc-sec; sea masked and shown flat), with the 51 SMD-TR strong-motion stations of the window drawn on the same colour scale and marker shape giving the V_{S30} reference flag; (b) the corresponding EC8/NEHRP site classes; (c) station V_{S30} against the proxy value sampled at the same coordinates, with the 1:1 line and factor-1.5 bounds. Values above 760 m s^{-1} saturate the colour scale. Station data: SMD-TR flatfile, NHERI DesignSafe PRJ-3950 v3, DOI 10.17603/ds2-f21x-s189, table Metadata.csv. Software used for producing graph: Python 3.12.3 with Matplotlib 3.9.3 (pyplot, colors.LightSource, ticker, patheffects), NumPy 2.4.6, SciPy 1.17.1 and rasterio 1.4.4. Data: GEBCO 2026 grid. Source: authors.